

Hybrid technique for optimal location and cost sizing of thyristor controlled series compensator to upgrade voltage stability

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Abstract: In this study, a hybrid technique is employed for ameliorating the voltage stability of test system with thyristor controlled series compensator (TCSC) installation. Improved gravitational search algorithm is used to first set the optimal location of TCSC to generate optimal power flow solution under normal and contingency condition while firefly algorithm is utilised to improve voltage stability and TCSC cost sizing, respectively. The proposed technique is implemented in MATLAB working platform and the power flow security and voltage stability is evaluated with IEEE 6, 14 and 30 bus transmission systems. The optimal TCSC placement is governed by new proposed index of apparent power flow index. The performance of the proposed method is deduced to be superior when results are compared with traditional gravitational search algorithm (GSA) and other hybrid methods of fuzzy-GSA and radial basis function neural network techniques.

1 Introduction

Planning and real time operation of power system is largely dependent on optimal power flow (OPF) and the emergence of competitive power market generates the obligation of minimising generation cost and system losses integrated with voltage stability. The OPF practically coordinates the power flow with gainful dispatch thus preparing for reduction in expense capacities by taking into record equality and inequality parameters [1, 2]. The problem of voltage instability is aggravated in application of open market principles where systems are generally operated close to their stability margins to attain maximum operational cost reduction. The traditional approach of Newton's strategy and direct programming are confronted with issues of being trapped in local minima, embraces losses and may not be adequately utilised to generate optimal solution [3–5]. Exceptional among the flexible AC transmission systems (FACTS) devices, thyristor controlled series compensator (TCSC) is invested with unfathomable application potential in unequivocally controlling the power flow on transmission line, damping power oscillations and venturing up transient robustness [6]. Numerous stochastic techniques like Genetic Algorithms (GA), Evolutionary Programming (EP) and Particle Swarm Optimisation (PSO), have been adequately utilised to effectively address the OPF problem with and without using FACTS controllers [7–9]. Nireekshana *et al.* [10] were instrumental in determination and change of available transfer capability in the deregulated working of power systems by static Var compensator and TCSC. Generally, most work stated in literature consider OPF and voltage stability issues independently but in open electricity markets it is imperative to consider voltage stability limits while accounting for profitable dispatch for overall system security.

The work delivered in this paper proposes a novel hybrid technique for optimising the location and cost sizing of TCSC resulting in secured OPF with retention of voltage consistency of the transmission system to prevent voltage collapse under variable load conditions. Two vital issues are considered: first is optimal TCSC placement for impregnable and reliable OPF solution under normal and contingency condition and second is to include voltage stability criteria in the original OPF objective function and TCSC cost reduction. The OPF problem is thus reformulated to elevate

emphasis on voltage stability. The remnant of the paper is presented as follows: Section 2 offers problem formulation followed by a detailed view of the proposed methods implemented. The results on three standard IEEE benchmark systems endorse the superiority of the proposed method.

2 Problem formulation

The TCSC is home to a sequence of compensating capacitors shunted by the thyristor controlled reactor. In this model, the controllable reactance is integrated in series to transmission line and the line impedance is adapted. As a result, the apparent power going through the line is either perked up to maximum or scaled down to minimum depending on loading conditions. The two objectives considered are secured OPF with optimal TCSC placement under normal and overloading state and evaluation of TCSC capacity in accordance with voltage variation and cost function.

Minimise fuel cost

$$F_1 = \sum (a_i + b_i P_{gi} + c_i P_{gi}^2) / h \quad (1)$$

Minimise

$$F_2 = (V_d, C_{TCSC}) \quad (2)$$

where a_i , b_i and c_i signify the fuel cost coefficients, V_d is the voltage deviation at the bus under overload condition and C_{TCSC} is the TCSC cost. The type and location of FACTS device implemented in the system influences the objective function and so firefly algorithm (FA) is employed to determine TCSC capacity in accordance with voltage variance and its cost function. The cost calculation of TCSC in US\$/kVar is based on Siemens AG Database [11] where s is the operating range of TCSC in Mvar

$$C_{TCSC} = 0.0015s^2 - 0.7130s + 153.75 \quad (3)$$

The power balancing conditions are equality constraints and limits on real and reactive powers generated; voltage magnitude and

TCSC reactance are taken as inequality constraints

$$P_{inj,i} = P_{g,i} - P_{L,i} \quad (4)$$

$$Q_{inj,i} = Q_{g,i} - Q_{L,i} \quad (5)$$

$$P_{g,i}^{min} \leq P_{g,i} \leq P_{g,i}^{max} \quad (6)$$

$$Q_{g,i}^{min} \leq Q_{g,i} \leq Q_{g,i}^{max} \quad (7)$$

$$V_i^{min} \leq V_i \leq V_i^{max} \quad (8)$$

$$-0.7X_{line} \leq X_{TCSC} \leq 0.2X_{line} \quad (9)$$

where $P_{inj,i}$ and $Q_{inj,i}$ are real and reactive powers injected into bus i , $P_{g,i}$ and $Q_{g,i}$ are real and reactive powers generated by i th generator and $P_{L,i}$ and $Q_{L,i}$ are real and reactive powers of i th load bus. V_i^{min} and V_i^{max} represent the lowest and highest voltage magnitude bounds of the i th bus, and X_{line} and X_{TCSC} characterise the line and TCSC reactance, respectively. To avert overcompensation, compensation range of TCSC is 20% inductive and 70% capacitive as given in (9) [12]. The underlying motive of the objective function is to initially ascertain optimal position of TCSC subject to OPF factors without infringing the equality and inequality constraints and also retaining voltage stability. The new index called apparent power flow index (API) is utilised for optimal TCSC installation for secured OPF

$$API = \frac{S_{i,Opf}}{S_{i,max}} \left(\frac{S_{i,L} - S_{i,n}}{S_{i,L} - S_{i,Opf}} \right) \quad (10)$$

where, $S_{i,n}$ is power flow in i th line at normal condition, $S_{i,Opf}$ is OPF in i th line, $S_{i,max}$ symbolises maximum power flow in i th line and $S_{i,L}$ is power flow of i th line under loading conditions. For placing TCSC optimally, the optimal line power flow is considered subject to maximum power flow limit and line flow variation of the system is analysed using API under varying load conditions. Overloading of transmission line in this case is set as 80% of normal load and added to the existing load based on which power flow calculations are done to obtain apparent power values. The API values are calculated for all the lines of the system considered and maximum value of API indicates the weakest bus and thus estimates the optimal location of TCSC. Then TCSC capacity is evaluated for TCSC cost sizing and minimum voltage deviation.

3 Computational techniques implementation

In this section, the intelligent and hybrid methods implemented to obtain optimal location and cost sizing of TCSC for OPF coupled with voltage stability are presented. The optimal location and

injected capacity of TCSC depend on the security indices of apparent power flow and cost function. The evaluation of effectiveness of each algorithm is analysed in two phases: first phase establishes optimal location of TCSC resulting in enhanced OPF and the second involves TCSC cost sizing and improving voltage profile. Subsequently, the injection capacity of TCSC is calculated by (8) and (9) through the security constraints. The IEEE standard bench mark 6, 14 and 30 bus systems are used for analysis and initially normal power flow and the stability conditions are determined and later the loading faults are introduced in the bus system. The detailed explanation of these computational methods is provided below.

3.1 Gravitational search algorithm

The gravitational search algorithm (GSA) has astonishingly arrived as one of the rewarding stochastic search techniques rooted on Newtonian laws of gravity and mass interaction [13]. In GSA, efficiency of the agents is evaluated by their masses which are deemed as objectives. The gravitational forces of attraction between the objects trigger a complete motion towards the objects which possesses heavier masses that procure superior fitness values and the best solution is approached as against the lesser ones which signify the worst solution. Each and every mass has the position in the order such as inertial mass, active gravitational mass and passive gravitational mass. The mass position amounts to the solution of objective function and the fitness function employed is indicated by the gravitational and the initial masses [14]. Traditional GSA method has low convergence due to increased randomness associated with agent's velocity which is suitably curbed by various hybrid techniques being integrated with it. Hybrid methods based on traditional GSA algorithm which are implemented in the problem to enhance the system performance are as follows.

3.2 Fuzzy-GSA

In classical GSA, agent's movements are governed by velocity equation in search space of the problem. However, velocity equation of the agents consists of random variables which lead the solution to uncertainties. Sequentially, to manage these unpredictability in solutions and to keep away detonation and deviation, velocity constraint is introduced which is predicted by fuzzy logic. The inputs of fuzzy system are current velocity of agents $v_i^d(t)$ and acceleration of agents $a_i^d(t)$ where 'genfis2' function is used to generate the fuzzy inference system.

For IEEE 6 bus system the graphical representation of membership functions for both the inputs of the fuzzy system are depicted in Fig. 1.

The block diagram of fuzzy logic controller with respect to input and output is shown in Fig 2.

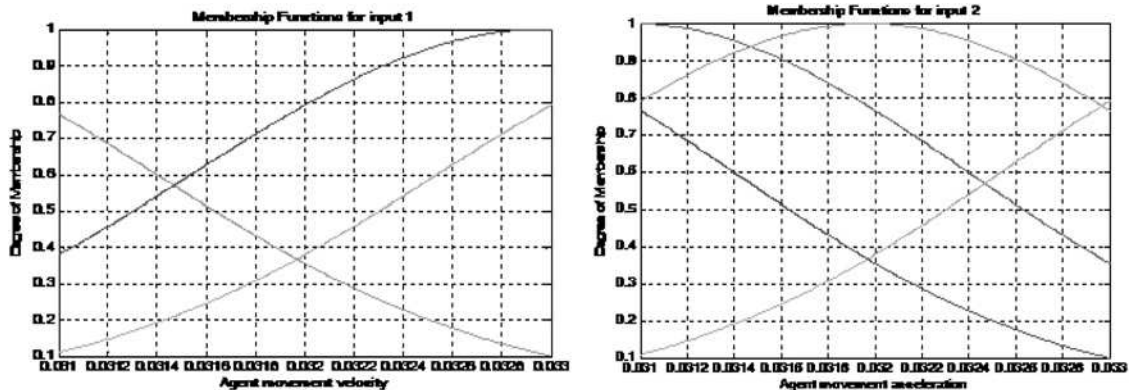


Fig. 1 Membership function for fuzzy system inputs

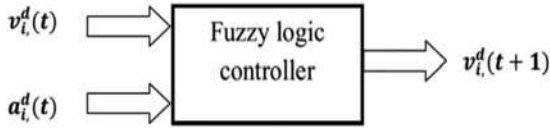


Fig. 2 Structure of fuzzy system

3.3 Radial basis function neural network (RBFNN)-GSA

In RBFNN-GSA, the velocity constraint to substantially reduce deviation is predicted by RBFNN instead of fuzzy logic as done earlier. The working procedure consists of three basic steps of initialisation, fitness evaluation and generating an updating function. In this regard, RBFNN is unique version of neural network encompassing three layers like input layer, hidden layer and output layer. The hidden nodes perform a set of radial basis function and the output nodes carry out linear summation functions as in multilayer perceptron [15]. Now, the RBFNN is entrusted with the function of allocation of random number in velocity updating equation to evaluate the new output velocity $v_i^d(t+1)$ where inputs consist of current velocity of agents $v_i^d(t)$ and agent's acceleration $a_i^d(t)$ and $x_i^d(t+1)$ is position of i th agent at time $t+1$. The function 'newrbf' is used to generate radial basis networks

$$x_i^d(t+1) = x_i^d(t) + v_i^d(t+1) \quad (11)$$

$$v_i^d(t+1) = \text{rand}_i \{v_i^d(t) + a_i^d(t)\} \quad (12)$$

3.4 Proposed method of improved GSA (IGSA) with FA

3.4.1 Phase 1-IGSA for optimum location of TCSC: IGSA is used for finding the optimum TCSC location by determining maximum API bus. The inputs are initialised as agents that represent the bus data. The mass of i th agent $m_i(t)$ and total mass of the agents $M_i(t)$ as estimated by mean is given by

$$m_i(t) = \frac{\text{fit}_i(t) - F_{\text{worst}}(t)}{F_{\text{best}}(t) - F_{\text{worst}}(t)} \quad (13)$$

$$M_i(t) = \frac{m_i(t)}{\sum_{j=1}^N m_j(t)} \quad (14)$$

where, $\sum_{j=1}^N m_j(t)$ represents the mass of total agents in s th search space, $\text{fit}_i(t)$ corresponds to fitness value of i th agent at instant t and $F_{\text{best}}(t)$ and $F_{\text{worst}}(t)$ represent the best and worst fitness of all agents. The parameters such as gravitational constant $G(t)$, $F_{\text{best}}(t)$, $F_{\text{worst}}(t)$ and $M_i(t)$ are rationalised. The gravitational constant $G(t)$ is initialised at the commencement, which paves the way for decreased convergence time and regulates search precision. The total force from heavier masses is applied and is evaluated in accordance with law of gravity such that acceleration of the i th agent and direction d at time t are chosen in accordance with law of motion and is given by

$$a_i^d(t) = \frac{F_i^d(t)}{M_i(t)} \quad (15)$$

The velocity variation of the searching technique for i th agent, direction d at time $t+1$ is expressed in (16) and $V_{\text{max}}^d(i)$ is the highest velocity

$$V_{\text{max}}^d(i) = V_{\text{max}}^0 * \psi_i \quad (16)$$

where

$$\psi_i = 1 - \left(\frac{(t_0)^h}{(t_{\text{max}})^h} \right), \quad V_{\text{max}}^0 = \eta \times (\delta_{\text{max}} - \delta_{\text{min}}), \quad (17)$$

The position of the agent is adapted as per (18)

$$\delta_{\text{max}} = \delta_i^d(t) + V_{\text{max}}^d(i) \quad (18)$$

The velocity of agents is in the range of $-V_{\text{max}} \leq V_{i(t+1)}^d \leq V_{\text{max}}$ in traditional GSA. This purposeless agent's adaption furnishes complication in the course of evaluation procedure, thereby turning out inaccurate solutions. In the novel technique proposed, velocity and position of agent search technique is enhanced based on the solution.

3.4.2 Phase 2-utilising FA for TCSC cost sizing and reduction in voltage variance: The FA represents a nature-inspired optimisation technique mainly rooted on behaviour of the flashing traits of fireflies where attractiveness is directly proportional to brightness or light intensity and both these qualities tend to decline as their distance goes on increasing. Here, the light intensity (I) is estimated to obtain fitness function which leads to minimisation of voltage variation between usual bus voltage and fault time bus voltage thus preserving the voltage consistency of the transmission system. The FA consists of initialisation process, fitness evaluation and update to get best solution. The population of fireflies is arbitrarily produced. Here, the arbitrary populations are deemed as voltages in the search space. The deviation in normal voltage (V_n) and fault time voltage is effectively estimated. The fitness function is estimated by the equations illustrated below. The brightness or light intensity of a firefly is affected or decided by landscape of the objective function to be optimised [16]

$$I = F(X_i) = \min(V_d, \cos t)$$

where

$$V_d = [V_n - V_i]_{i=1}^N \quad (19)$$

The algorithm is updated such that from initial fitness value the least minimum fitness is chosen and that gives quality of best solutions. The subsequent cycle of firefly commences with generation of new solution X_j having fitness function F_j . At this juncture, we verify the condition ($F_j < F_i$), and if this stipulation is met, then the solution is shifted else the attractiveness of the fireflies has to be revised and fireflies will be shifted to their new location as given by equation below where α represents the maximum radius of the arbitrary step and β corresponds to attractiveness of fireflies which decides the step size in the direction of optimal solution (see Table 1)

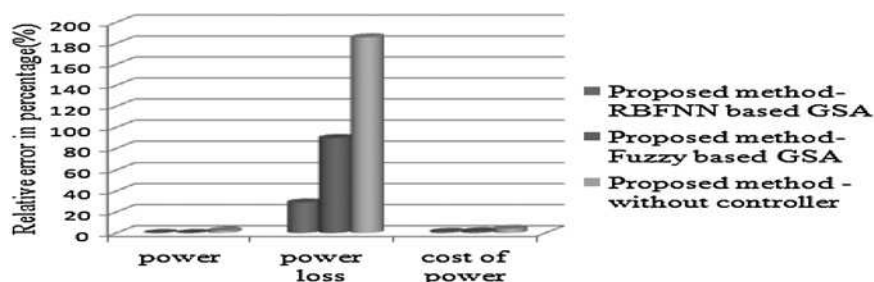
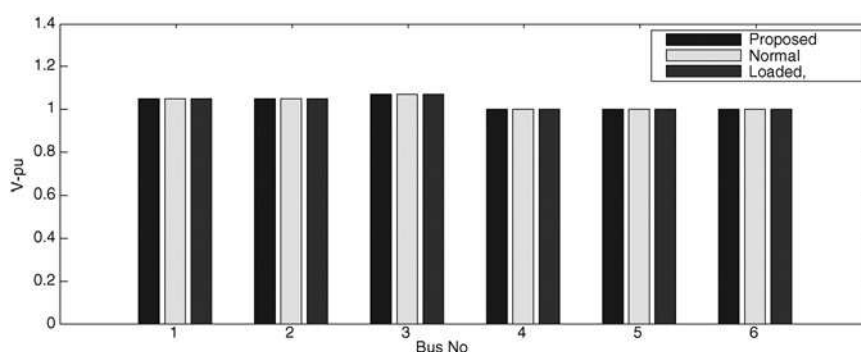
$$X_i^{t+1} = X_i^t + \beta(x_j - x_i) + \alpha \left(\text{rand}() - \frac{1}{2} \right) \quad (20)$$

Table 1 Implementation parameters

S.No	Description	Algorithms	Values
1	maximum generation	FA	50
2	n		20
3	alpha		1
4	beta min	IGSA	0.3
5	gamma		0.5
6	maximum iteration		50
7	n		20
8	gravitational constant		13.5335
9	alpha		20
10	G0		100

Table 2 Secured OPF with TCSC cost sizing for IEEE 6 bus system

S.No	Total active power generation in MW					
1	loading condition	TCSC in line between 5 and 6 buses with different methods				
		GSA	fuzzy-GSA	RBFNN-GSA	proposed IGSA-FA	
1	prior to loading	210			210	
	subsequent to loading without TCSC	215				
2	total real power loss in MW					
	prior to loading	6.5393			3.4213	
	subsequent to loading without TCSC	8.48				
3	total active power generation cost, \$/hr					
	prior to loading	3460.2349			3013.3921	
	subsequent to loading without TCSC	3920.1708				
4	TCSC cost, \$/MVAR					
		162.76	153.8921	148.7653	119.0594	

**Fig. 3** Comparison performances of OPF with different methods**Fig. 4** Performance analysis of bus voltage using proposed method

4 Results and discussion

In this paper, innovative hybrid techniques based on GSA is executed in MATLAB platform and their performances are evaluated on IEEE standard bench mark system of 6 and 14 buses, respectively. Among all the methods implemented in the test systems, IGSA-FA computational technique provides superior results. TCSC installation elevates the system performance to achieve dual intent of OPF coupled with voltage stability and effectiveness is assessed by random overloading of transmission lines to ascertain best TCSC location and secured reactance limit. The bus data, line data and the limits of control variables are estimated from [17, 18]. The fuel cost coefficient of IEEE 14 bus system is referred from [19]. The Newton–Raphson power flow method is effectively employed to calculate the power flow solution before and after setting TCSC. The test systems are evaluated using traditional GSA, fuzzy-GSA, RBFNN-GSA and the proposed method wherein outcomes are assessed and contrasted in each case.

4.1 Case 1: analysis of IEEE 6 bus system

This section describes the effectiveness of the proposed hybrid technique with IEEE 6 bus system where total active power generation, real power losses, generation cost and TCSC cost are evaluated and results are tabulated in Table 2. A comparative

account with other methods dictates the superior performance of the proposed method of IGSA-FA. The total cost of active power generation and real power losses with the proposed method is less than that obtained in [20] for case of OPF with TCSC both under normal and contingency conditions. For the 6 bus system, location of TCSC connecting line is optimised between 5 and 6 buses and optimal TCSC placement appreciably reduces the overload of transmission line.

Fig. 3 below depicts the statistical analysis of individual relative error (%) which coherently establishes the proposed method to be the pre-eminent method among all the hybrid techniques. The formula for individual relative error (%) is illustrated as

$$\text{Individual relative error (\%)} = \frac{(\text{TCSC with othercases} - \text{TCSC with IGSA-FA})}{\text{TCSC with IGSA-FA}} \times 100 \quad (21)$$

Fig. 4 illustrates the voltage profile variation of the IEEE 6 bus system with IGSA-GA and placing TCSC optimally paves the way for secured power flow and revamps voltage stability. The voltage profile under normal condition and loaded condition with the proposed method is captured and it is evident that the voltage collapse even under inconsistent loading is prevented while OPF is established.

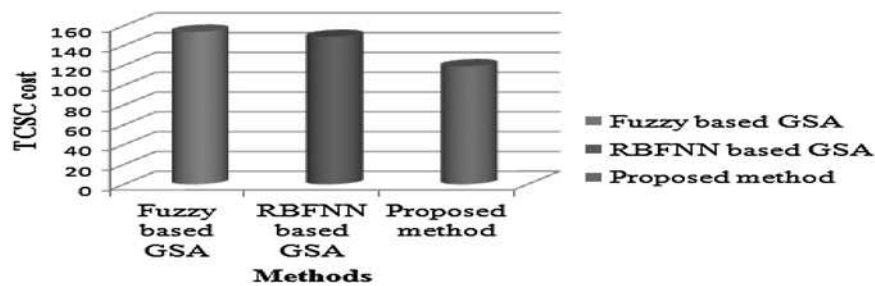


Fig. 5 Analysis of TCSC cost with different methods

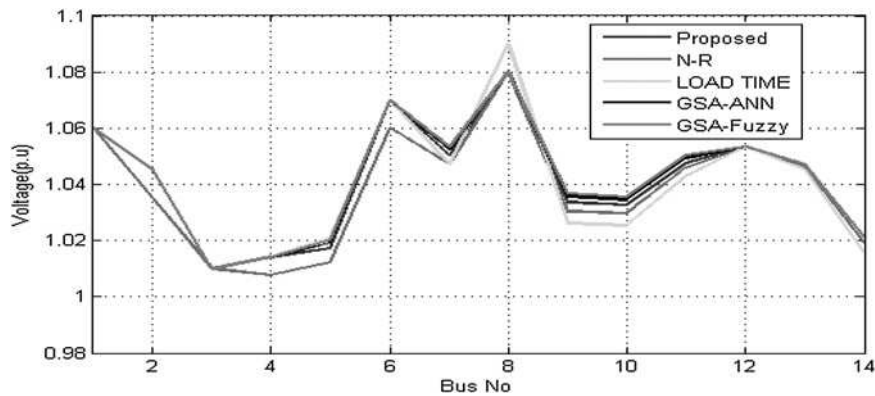


Fig. 6 Comparison analysis of bus voltages with various methods

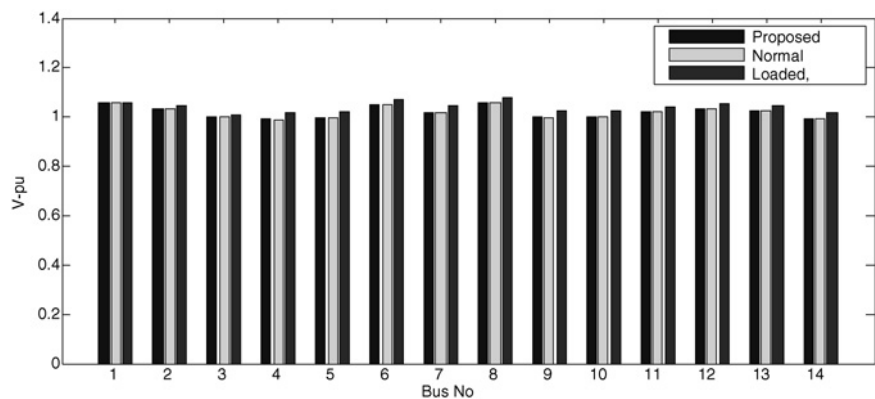


Fig. 7 Comparison analysis of bus voltages with various methods

Fig. 5 illustrates that system performance based on TCSC cost sizing with the proposed method being compared with other existing methods. The TCSC cost obtained by the proposed method, RBFNN-based GSA and fuzzy-based GSA is 119.0594, 148.7653 and 153.8921, respectively. The deviation rate of cost reduction in the proposed method is 34.8327 and variation rate is about 29.7059–34.8327. Therefore, the proposed method is utilised to reduce the TCSC cost substantially when compared with other techniques.

4.2 Case 2: analysis of IEEE 14 bus system

The effectiveness of the proposed technique is tested in IEEE 14 bus system where initially the power flow is identified to establish enhanced OPF and subsequently voltage stability is analysed using the proposed method. The voltage profile of the system is analysed and compared with existing methods which is illustrated in Figs. 6 and 7. It shows that, the proposed method effectively maintains the voltage profile within the mentioned limit (1.02–1.0 p.u.) even under variation of loading condition.

The secured OPF results with diverse hybrid methods implemented and TCSC cost sizing is analysed and represented in Table 3. The real power losses with the proposed method are substantially reduced when compared with [20].

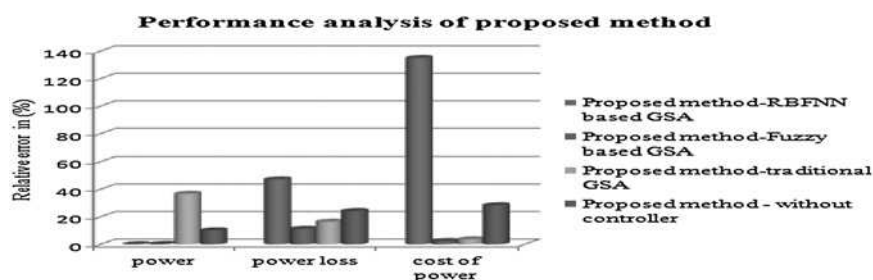
The comparison chart of IEEE 14 bus individual relative error (%) and for cases without TCSC and with TCSC installed implementing GSA, fuzzy-GSA and RBFNN-GSA is illustrated in Fig. 8 and tabulated in Table 4. Also, from Table 3 the TCSC cost is 117.5868 \$/MVAR which is significantly reduced.

4.3 Case 3: analysis of IEEE 30 bus system

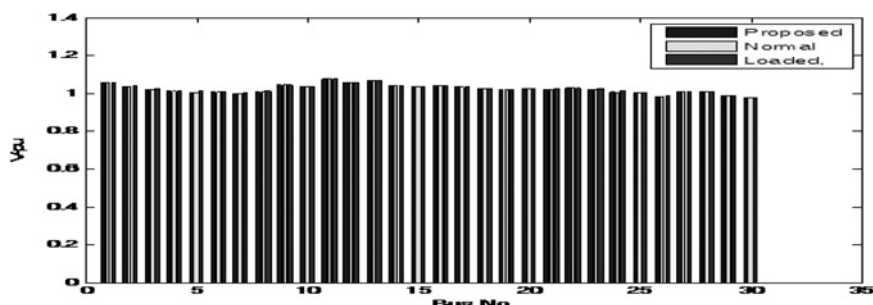
The effectiveness of IGSA-FA algorithm is validated on IEEE 30 bus system for secured OPF profile. It consists of 6 generators, 41 lines and cost coefficient values are tabulated below. The voltage profile in Fig. 9 under normal and loaded condition with the proposed method depicts retention of bus voltage stability. A comparison of simulation result with respect to first objective of problem formulation to minimise fuel cost, real power losses for secured OPF considering IEEE 30 bus system is given in Table 5.

Table 3 Active power profile with respect to secured OPF

S.No	Total active power generation in MW					
1	loading condition	TCSC in line of between 5 and 6 buses with different method				
		GSA	fuzzy-GSA	RBFNN-GSA	IGSA-FA	
	prior to loading	258.3396	259.9853	259	258.4563	
	subsequent to loading without TCSC	285.55	259.9539	259		
2	total real power loss in MW					
	prior to loading	6.5929	7.95	7.6	6.8227	4.6322
	subsequent to loading without TCSC	8.48				
3	total active power generation cost, \$/hr					
	prior to loading	3460.2349	3164.4385	3125.9924	3050.1011	1297.4326
	subsequent to loading without TCSC	3920.1708				
4	TCSC cost, \$/MVAR		153.6841	145.3623	132.6382	117.5868

**Fig. 8** Performance analysis of proposed method with existing method in IEEE 14 bus system**Table 4** Individual relative error of proposed method with other techniques

Methods comparison	IGSAFA-RBFNN GSA	IGSAFA-Fuzzy GSA	IGSAFA-traditional GSA	IGSAFA-without TCSC
power	0.210364383	0.368301158	36.8301158	10.25096525
power loss	47.2885454	11.39285034	16.52278423	24.29096985
cost of power	135.0874412	2.488156868	3.748642955	28.52592985

**Fig. 9** Bus voltage profile during normal, loaded case with proposed technique**Table 5** Comparison of simulation result for secured OPF with fuel cost functions for IEEE 30 bus system

Methods	Minimum fuel cost, \$/h	Real power losses, MW	CPU, s
IGSA-FA	795.675	2.6351	10.045
BBO [21]	799.1116	-	-
EP [22]	802.5557	9.3683	51.4
PSO [23]	802.0136	9.3301	77.672
ACO [24]	803.0123	9.4616	20
ABC [25]	800.1874	8.9197	22.73

Table 6 Fuel cost coefficients of IEEE 30 bus and real power limits

Buses	$P_{G,min}$, MW	$P_{G,max}$, MW	a_i	b_i	c_i
generator					
1	50	200	0	2	0.0038
2	20	80	0	1.75	0.0175
5	15	50	0	1	0.0625
8	10	35	0	3.25	0.0083
11	10	30	0	3	0.0250
13	12	40	0	3	0.0250

The result comparison shows that the proposed technique yields better result with enhanced computational speed as evident from CPU time comparison. Table 6 provides the system data for IEEE 30 bus system.

Table 7 depicts the effective reduction in mean CPU time by proposed method where the results are averaged over 30 runs to attain the best solution. The processor configuration on which the

Table 7 Comparison of mean CPU time for various computational techniques

Computational methods	Type of IEEE system	Mean CPU time, s
GSA	6 bus transmission system	12.184994
fuzzy-GSA		10.623367
RBFNN-GSA		10.26114
IGSA-FA		2.68646
GSA	14 bus transmission system	14.120289
fuzzy-GSA		12.612512
RBFNN-GSA		11.76395
IGSA-FA		4.782955

algorithms are executed is: Intel (R) Core 2 duo CPU, 1.83 GHz, 2.99 GB RAM running on Windows XP and is coded in MATLAB R2008b.

5 Conclusion

The work presented in this paper emphasises on achieving secured OPF but without compromising voltage stability even under heavy loading condition and candidly displays four approaches of GSA, fuzzy-GSA, RBFNN-GSA and IGSA-FA to achieve the same. The proposed method of IGSA-FA generates superlative results with optimised TCSC location providing minimum generation cost, TCSC cost and diminished power losses for standard IEEE benchmark 6 and 14 bus systems. The power flow security is suitably analysed by each method as line outage reduces the load power limits which are restored eventually by optimally setting the installed TCSC to rebuild secured power flow of transmission lines and retain stable voltage profile. The result comparison for IEEE 30 bus system for OPF with respect to cost functions reveals the superior performance of the proposed algorithm for better optimal solution with enhanced computational speed. The comparison among all the methods by graphical representation, results and statistical analysis of Individual relative error (%) manifest the escalated performance of the proposed method.

6 Acknowledgment

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